Memoization

i member

¿Cuál es el peligro de la Recursión?

Experimentémoslo empíricamente:

```
long fibonacci(int pos) {
    if( pos == 0 ) return 0;
    if( pos == 1 ) return 1;
    return fibonacci(pos - 1) + fibonacci(pos - 2);
}
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A penas 50!!

Análisis formal

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}
```
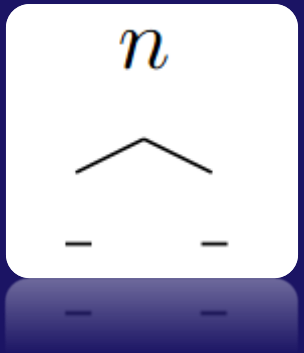
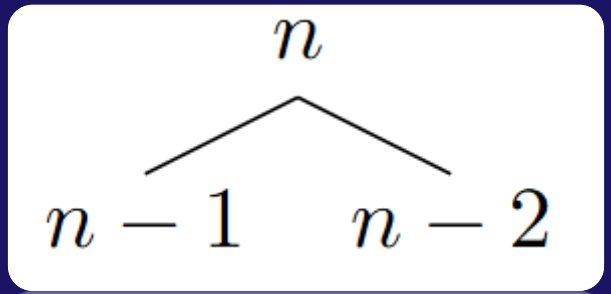
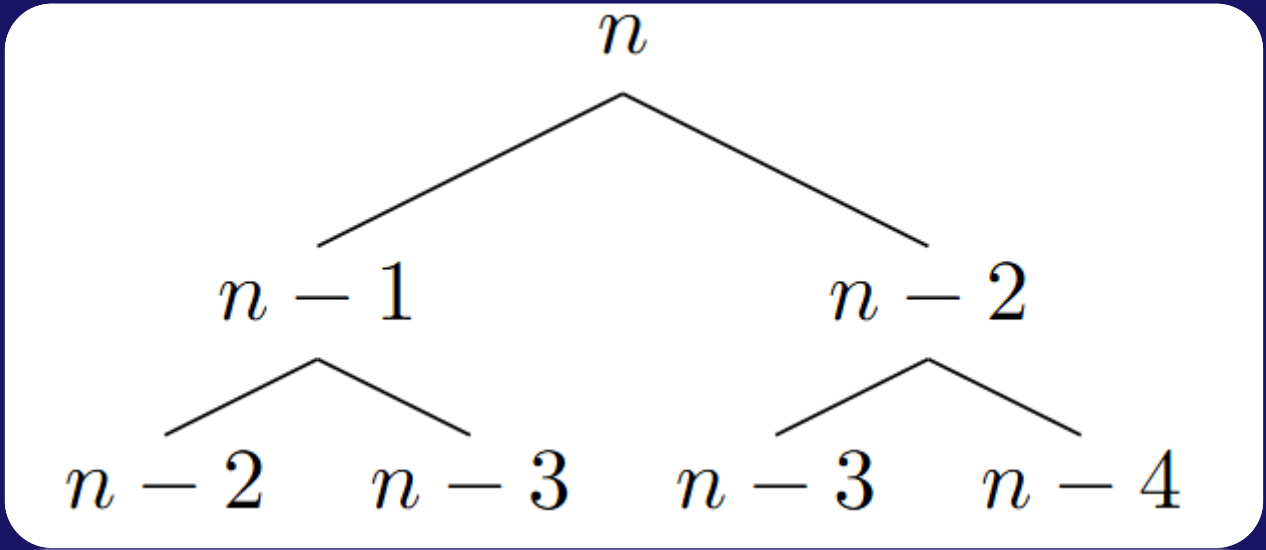

Recurrencia:

$$T(n) = T(n-1) + T(n-2) + O(1)$$

$$n$$


n


n
$n-1$
$n-2$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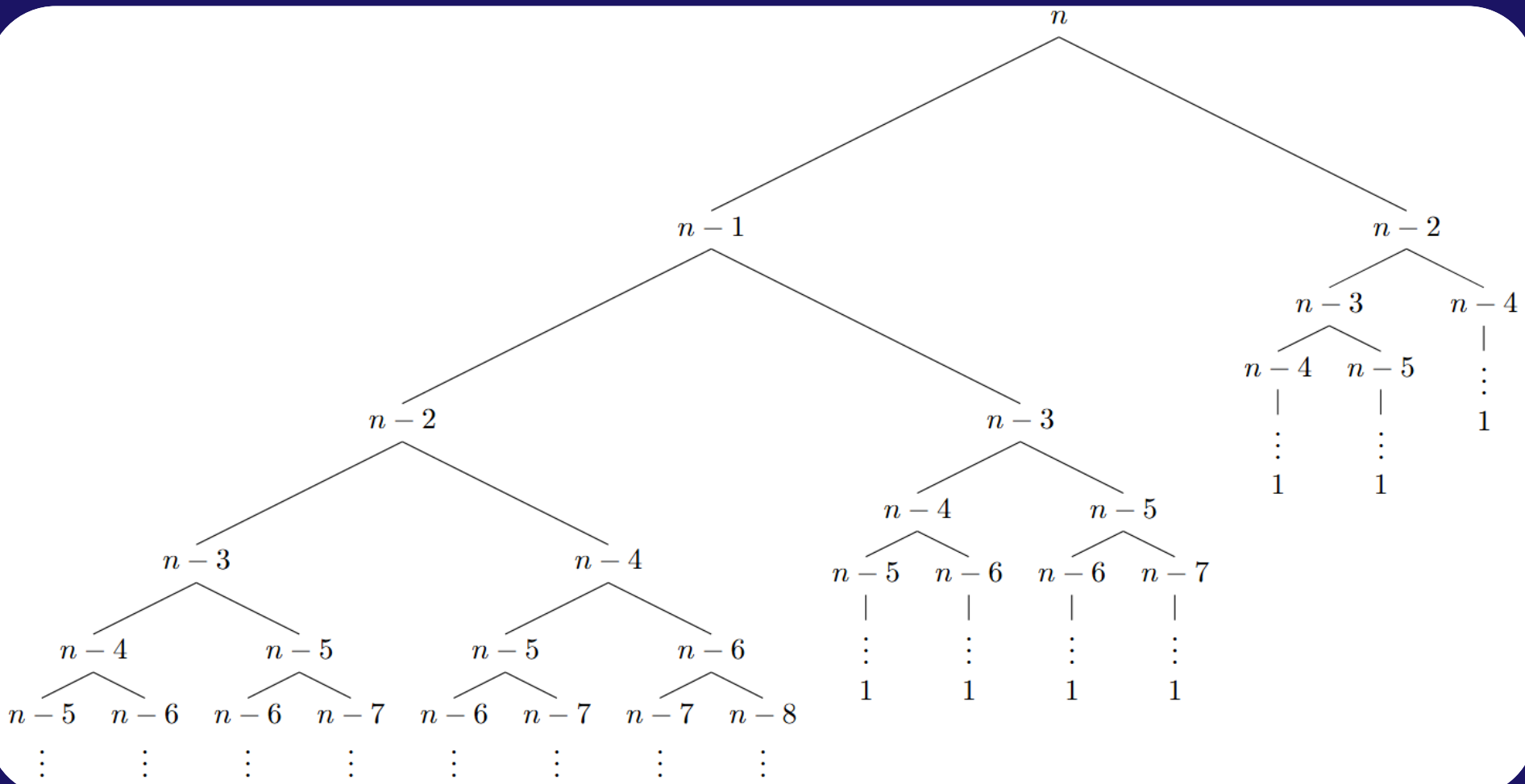
$n-4$


n
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$n-7$
$n-6$
$n-7$
$n-7$
$n-8$
1
1
1
1
1
1
1
1
1

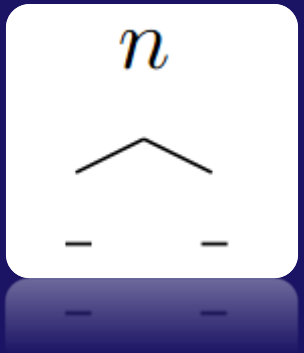
Análisis formal

```
long fibonacci(int pos) {
    if( pos == 0 ) return 0;
    if( pos == 1 ) return 1;
    return fibonacci(pos - 1) + fibonacci(pos - 2);
}
```

Recurrencia:

$$
\begin{aligned}
T(n) &= T(n-1) + T(n-2) + O(1) \\
&\leq T(n-1) + T(n-1) + O(1) \\
&= 2T(n-1) + O(1)
\end{aligned}
$$

$$n$$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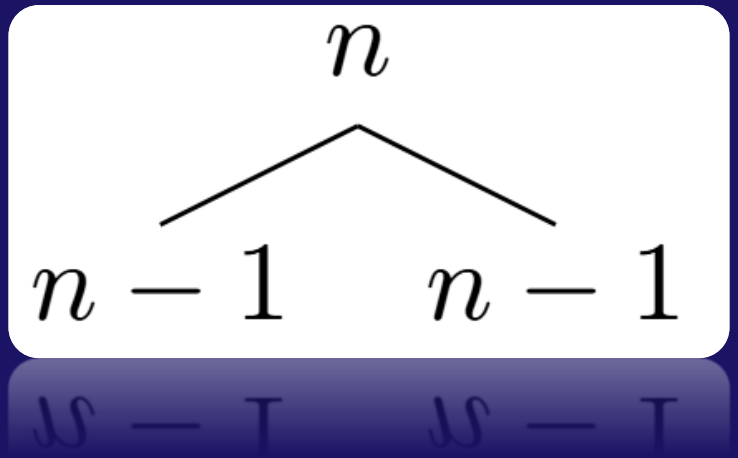
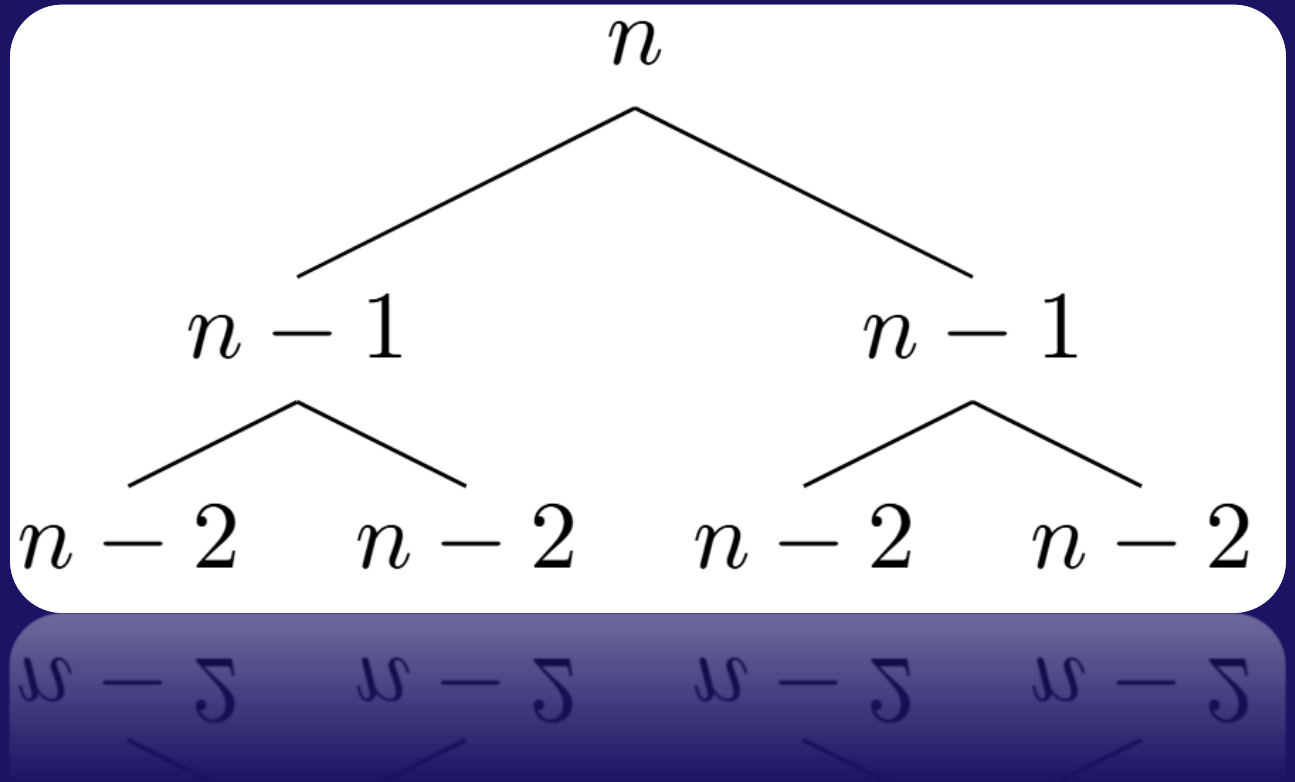

n
$n-1$
$n-1$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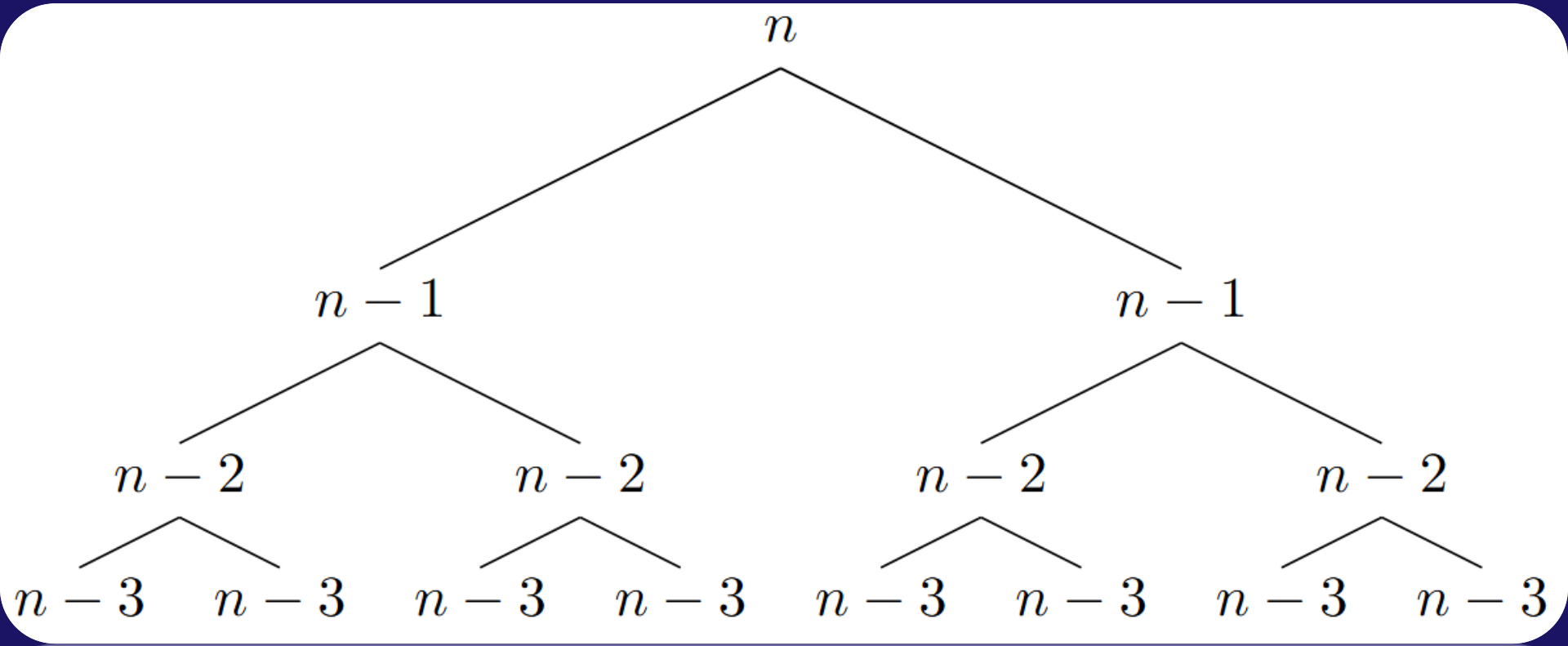
$n-2$
$n-2$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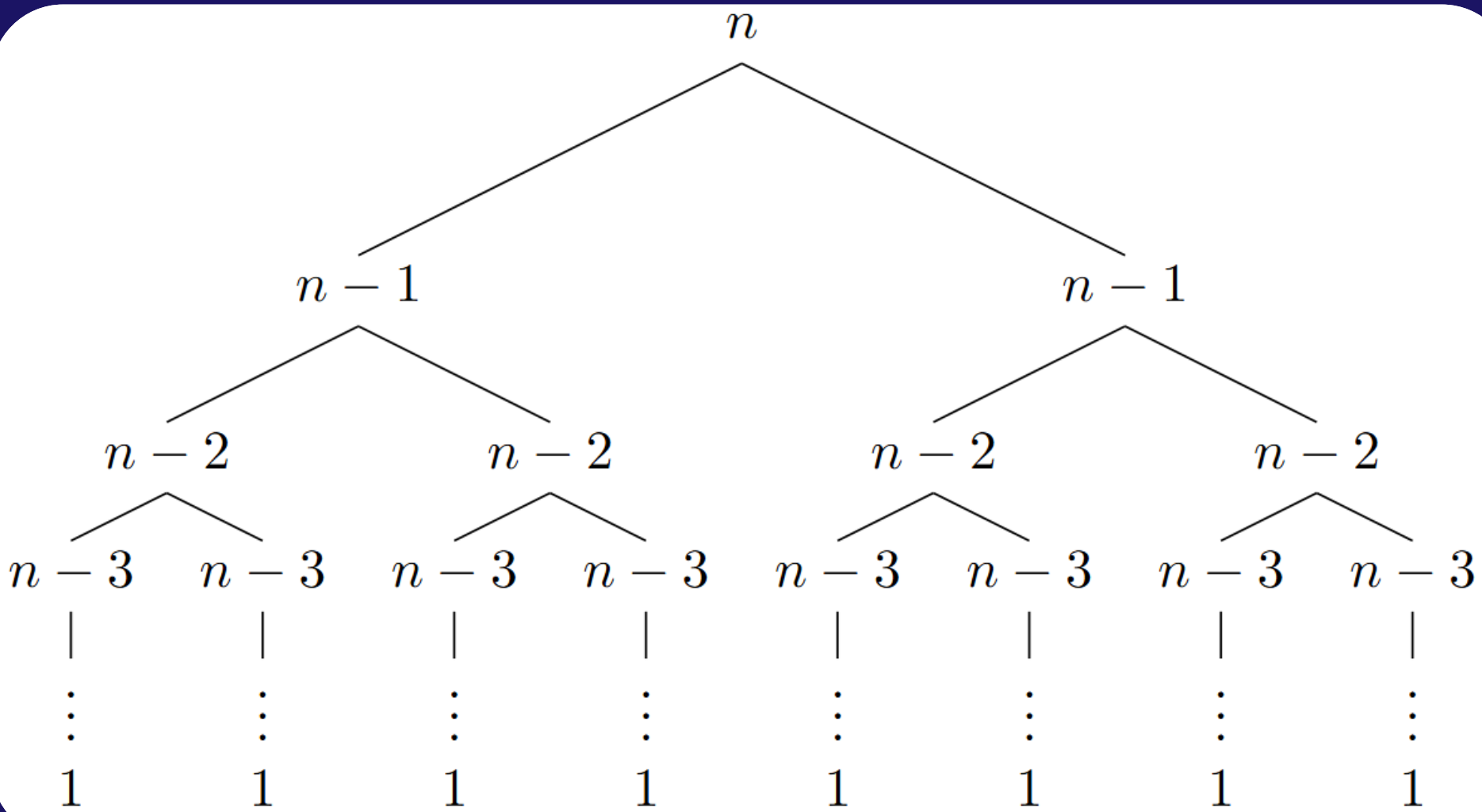
n
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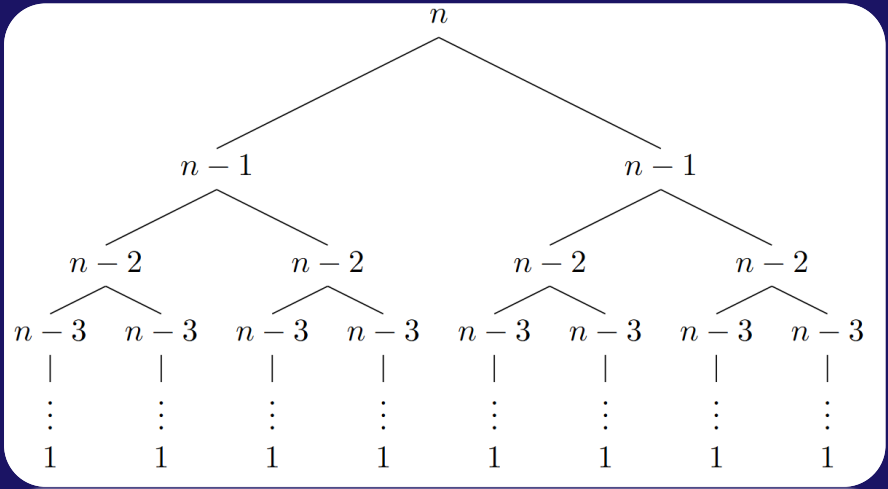
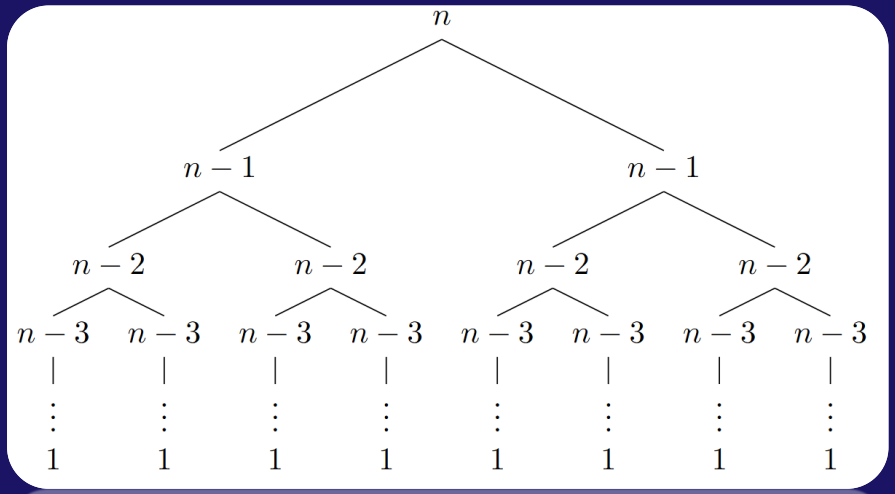
$$T(n) = 2T(n-1) + O(1)$$



1. Costo de un dato: c
2. Datos por nodo: $n - k$
3. Costo de nodo: $c\boldsymbol{f}(n-k) = c$
4. Nodos por nivel: 2^k

$$T(n) = 2T(n-1) + O(1)$$



5. Costo de nivel: $c(2^k)$

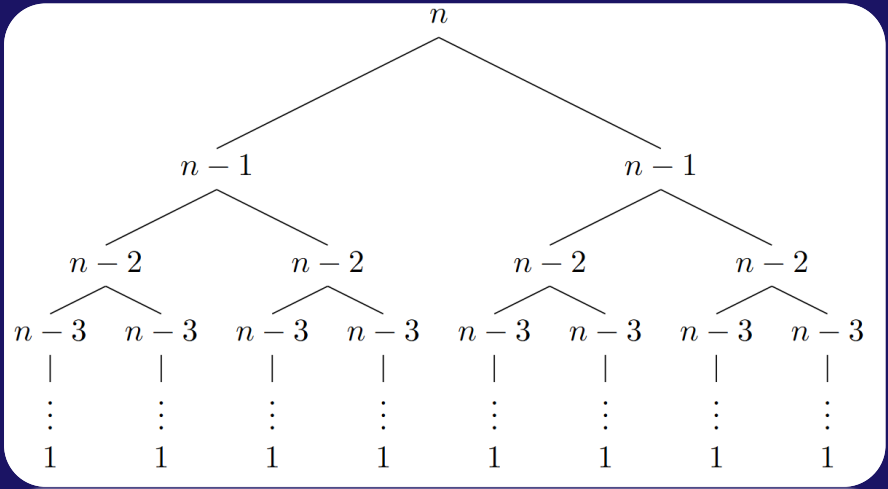
6. Costo de árbol:

$$\sum_{k=0}^{h} c(2^k) = c \sum_{k=0}^{h} 2^k$$

$$n - h = 1 \rightarrow h = n - 1$$

$$c \sum_{k=0}^{n-1} 2^k$$

$$T(n) = 2T(n-1) + O(1)$$



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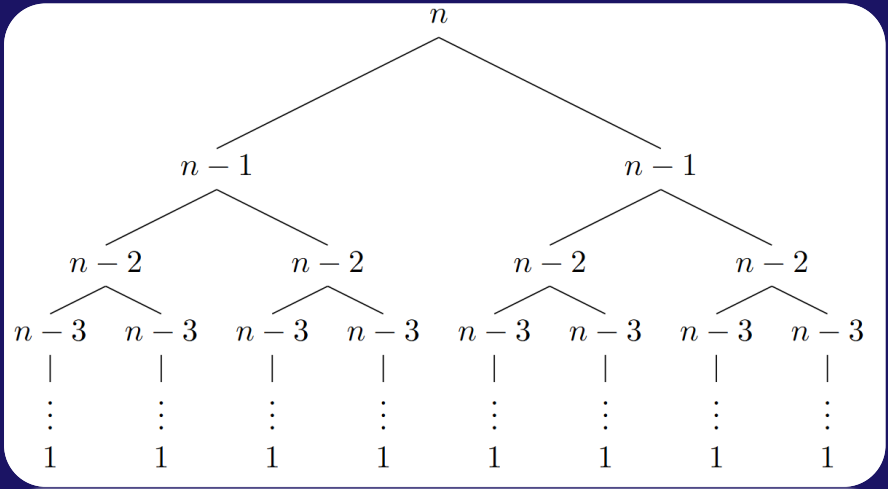
$$c \sum_{k=0}^{n-1} 2^k$$

$$\sum_{k=0}^{n-1} ar^k = \frac{a(1-r^n)}{1-r}$$

$$T(n) = 2T(n-1) + O(1)$$



5. Costo de nivel: $c(2^k)$

6. Costo de árbol:

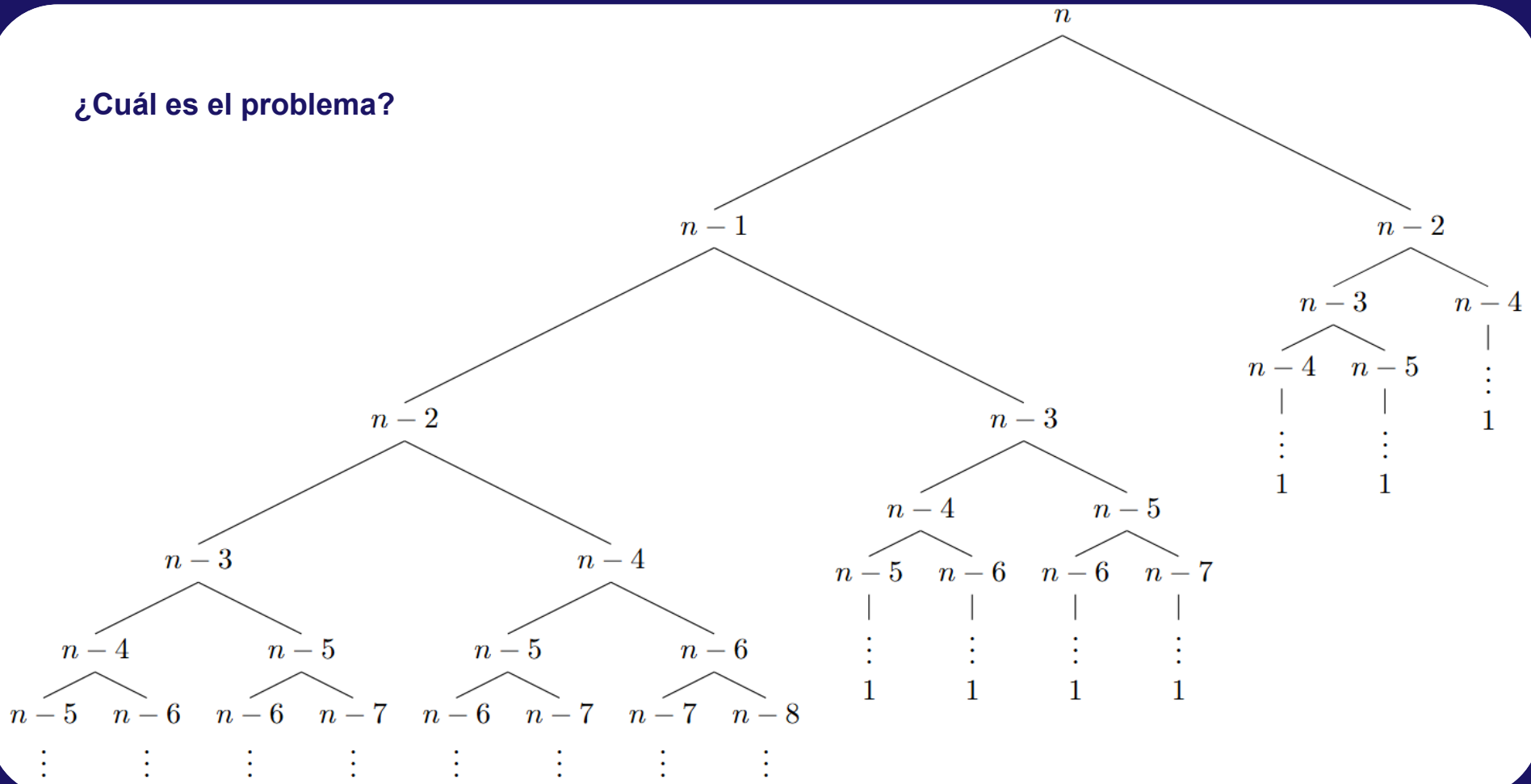
$$c\sum_{k=0}^{n-1} 2^k = c\frac{1-2^n}{1-2} = c(2^n - 1) = c(2^n) - c = O(2^n)$$

Contraste curioso:

```
int factorial_posposicion(int n) {
    if( n == 0 || n == 1 ) return 1;
    return n * factorial_posposicion(n-1);
}
```

$$T(n) = O(n)$$

¿Cuál es el problema?



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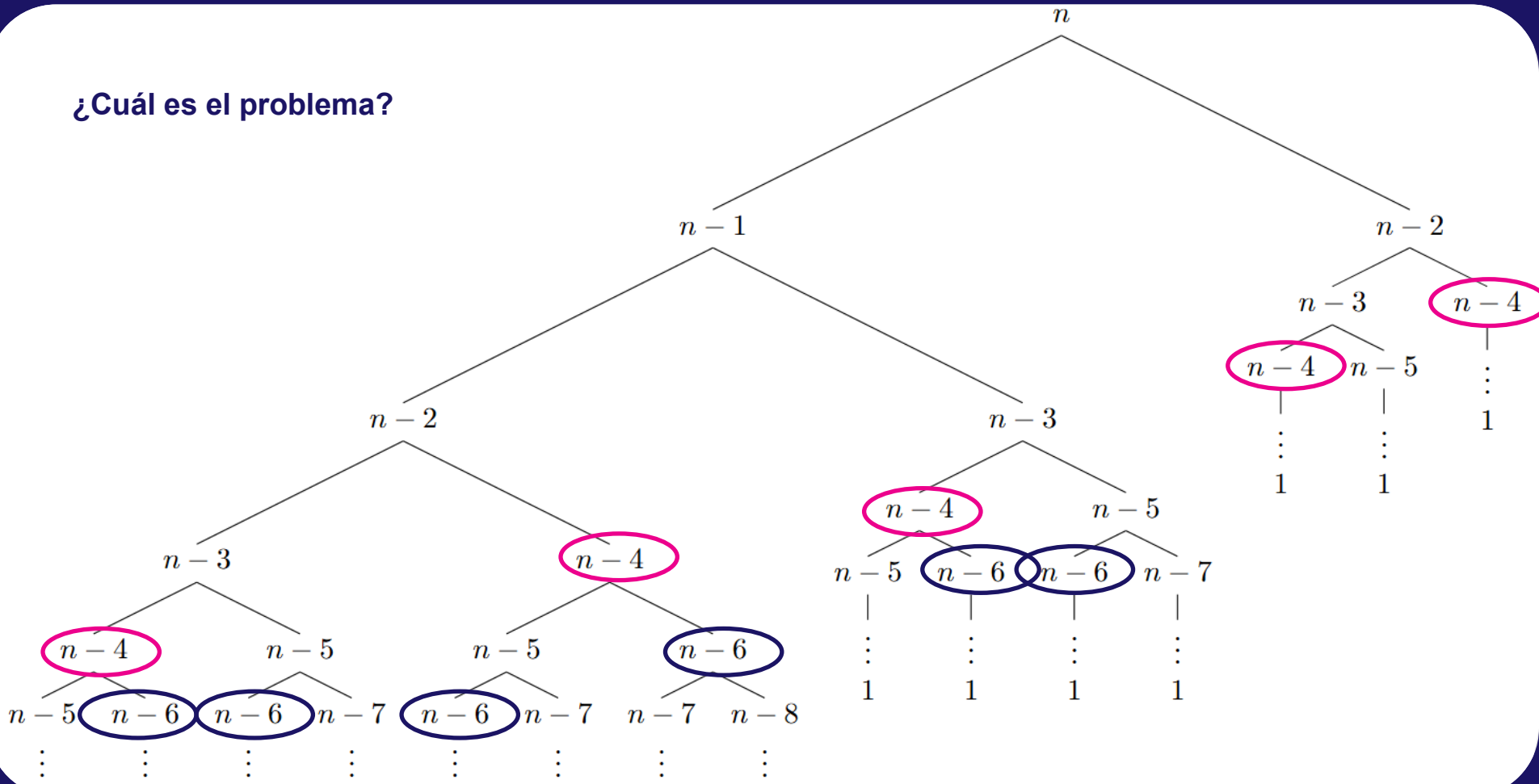
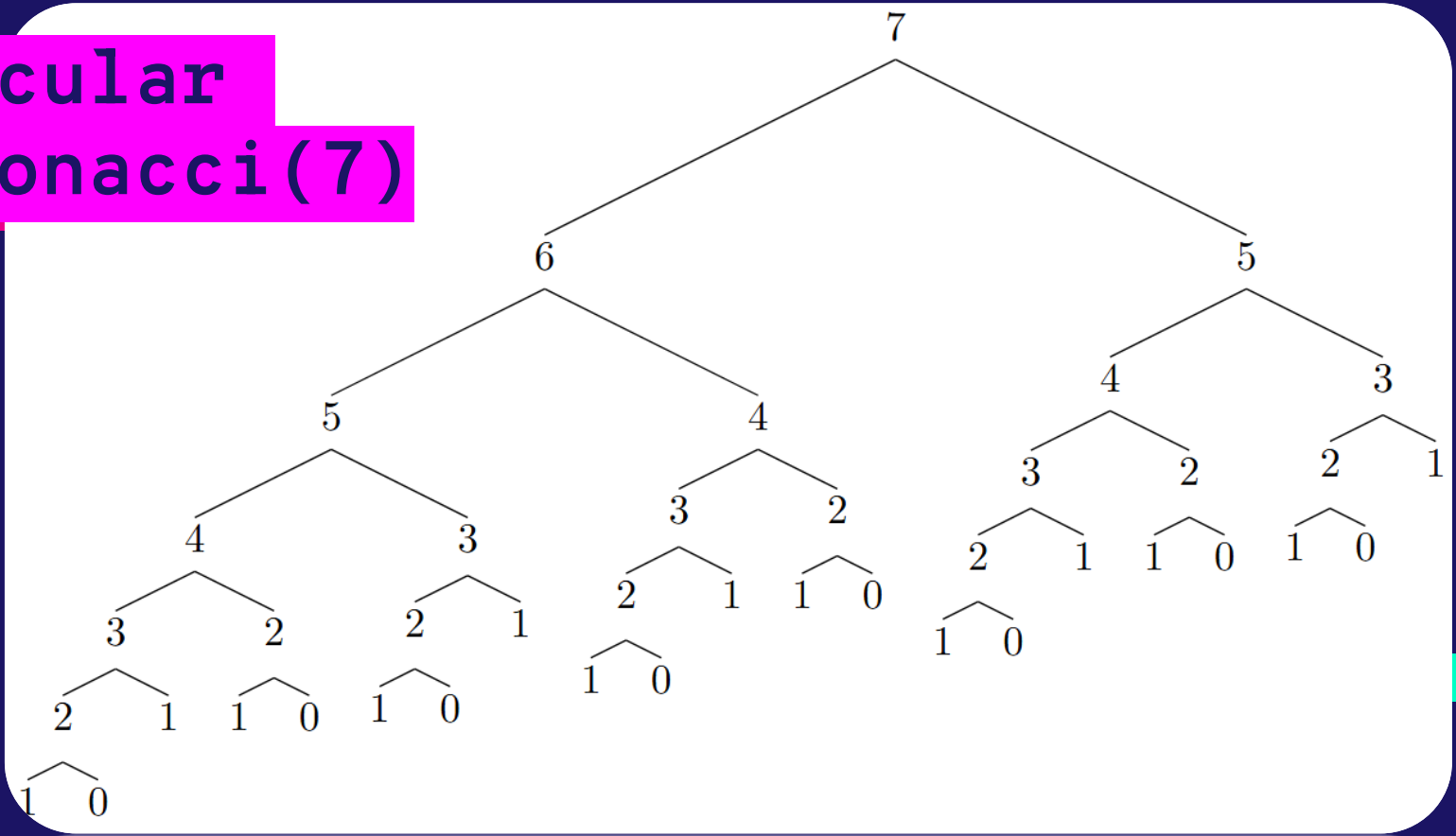


Calcular fibonacci(7)



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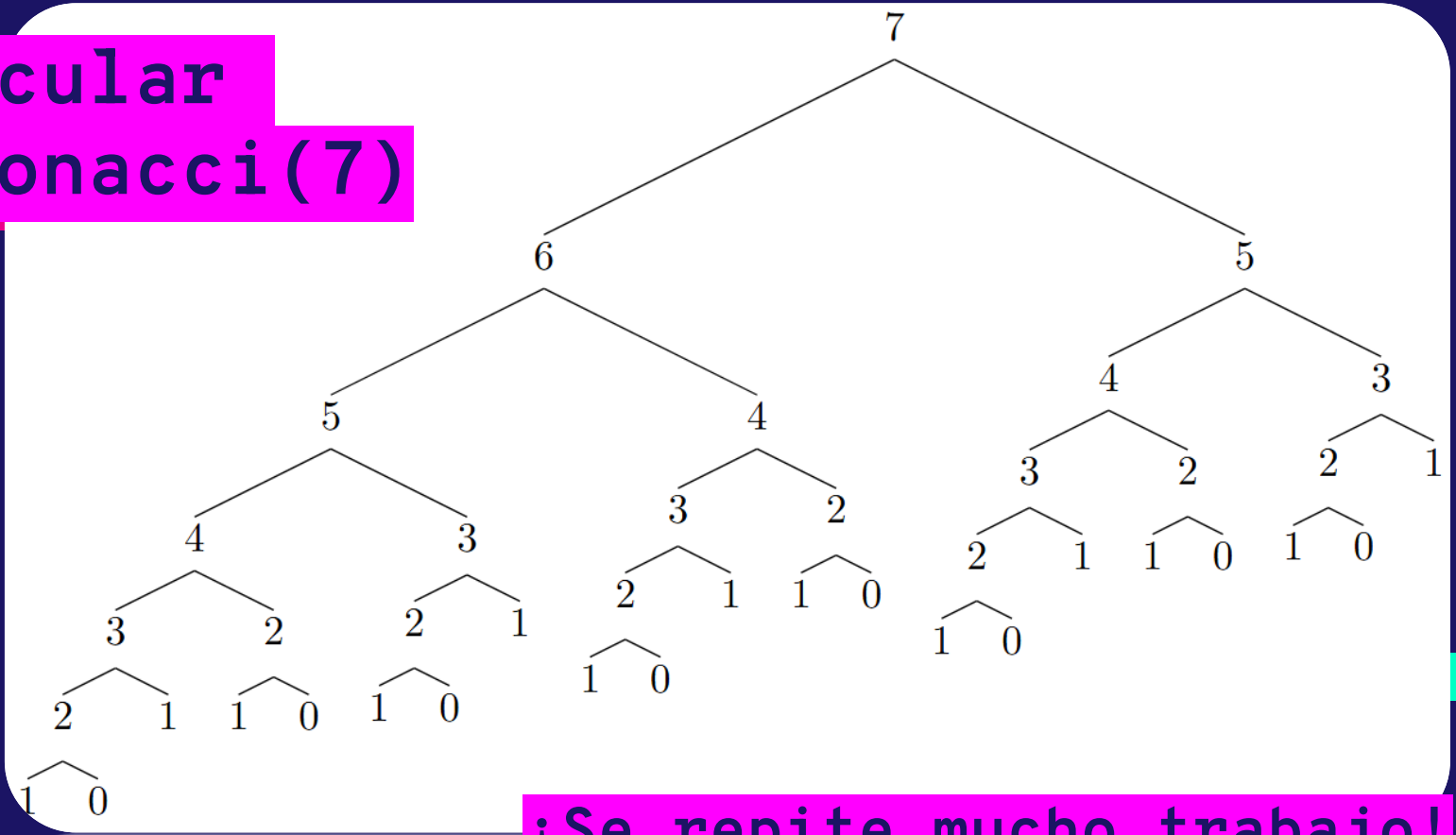


¡Se repite mucho trabajo!

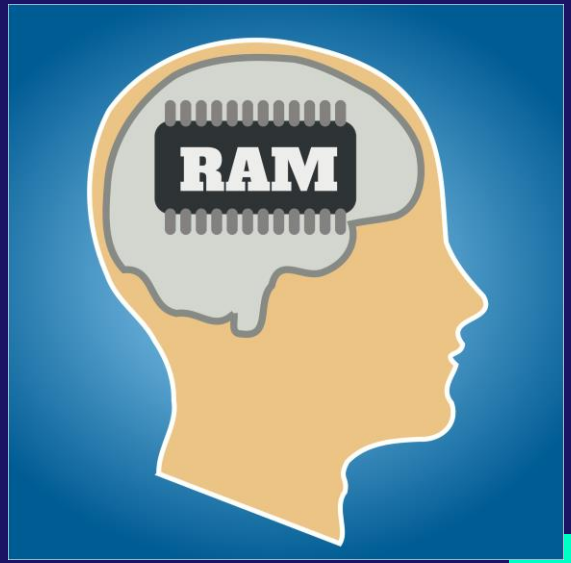
MEMOIZATION

Es una técnica de optimización utilizada para mejorar el rendimiento de los algoritmos al **almacenar los resultados de llamadas a funciones costosas** y reutilizarlos cuando vuelven a aparecer durante la ejecución.

Memoization

- La primera vez que una tarea ocurre:
  - Se calcula
  - Se almacena el resultado

- La segunda vez y veces posteriores:
  - Se reutiliza el resultado ya calculado



```
long fibonacci_memo(int pos) {
    if( pos == 0 ) return 0;
    if( pos == 1 ) return 1;

    vector<long> memo(pos+1,-1);
    memo[0] = 0;
    memo[1] = 1;

    return fibonacci_memo_aux(pos, memo);
}
```

```
long fibonacci_memo_aux(int pos, vector<long>& memo) {
    if( memo[pos] != -1 ) return memo[pos];
    memo[pos] = fibonacci_memo_aux(pos-1, memo) + fibonacci_memo_aux(pos-2, memo);
    return memo[pos];
}
```

Muchas Gracias

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